BUSINESS REQUIREMENTS DOCUMENT

BeOne POC

Feature Summaries
Prepared by Kolosal AI

Version 1.0 · October 20, 2026 · Draft · Internal

BUSINESS REQUIREMENTS DOCUMENT

01 - Onboarding

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Welcome Screen serves as the personalized landing page for users after they log in. It provides a professional, branded greeting while offering an immediate snapshot of the user's current responsibilities and system status. By displaying role-specific information and active tasks, the screen ensures users are oriented and ready to begin their work efficiently.

1.2 Step-by-Step Flow

1. Authentication Check: Upon arrival, the system verifies the user's identity. A loading screen with a spinner is displayed while this information is being retrieved.
2. Profile Display: Once authenticated, the screen displays a personalized greeting using the user's first name, accompanied by their profile avatar.
3. Role Recognition: The system highlights the user's primary and secondary roles using color-coded badges, making it clear what permissions the user holds.
4. Task Overview: The screen provides a quick summary of the user's current workload, specifically highlighting the number of drafts in progress and items currently awaiting their review.
5. Contextual Information: A side panel displays platform value propositions (such as compliance and speed metrics) alongside a disclaimer regarding administrative role management.
6. Navigation: The user can click the "Continue to Dashboard" button to proceed to their main workspace.

1.3 Key Concepts

- Role-Awareness: The interface dynamically adjusts to show specific job functions (primary and secondary roles), ensuring users understand their current access level.
- Task Summary: A high-level view of pending work, designed to help users prioritize their actions immediately upon logging in.
- Authentication State: The system manages the transition from a "loading" state to a "personalized" state, ensuring a smooth user experience during the login process.
- Compliance Disclaimer: A clear notification that role-based permissions are managed by administrators, providing transparency regarding system access.

1.4 Business Value

- Efficiency: By surfacing active tasks (drafts and reviews) immediately, users can jump directly into their most urgent work without searching through menus.

- User Confidence: The personalized greeting and clear display of roles reinforce that the user is in the correct environment with the appropriate permissions.
- Brand Alignment: The inclusion of platform performance metrics (e.g., compliance and speed) reinforces the value of the tool to the user every time they log in.
- Reduced Support Overhead: By explicitly stating that role permissions are managed by administrators, the system proactively answers common questions about access levels.

2. Logic

2.1 Overview

The Onboarding feature serves as the personalized "front door" for the BeOne platform. When a user logs in, the system automatically identifies who they are and what their responsibilities are. This ensures that the welcome screen is tailored specifically to the individual, displaying their name, professional role, and a quick snapshot of their current workload so they can immediately see what needs their attention.

2.2 Step-by-Step Flow

1. Loading State: As soon as the user initiates the login process, the screen displays a "Signing you in..." message with a loading spinner to indicate that the system is preparing their personalized workspace.
2. Profile Retrieval: The system securely fetches the user's identity details, including their name, profile picture, and assigned professional roles.
3. Task Summary Calculation: Simultaneously, the system scans the user's active tasks to count how many are currently in a draft state and how many are waiting for their review.
4. Personalized Dashboard Display: Once the data is ready, the screen updates to show a warm welcome message, visual badges representing the user's roles, and a summary of their pending tasks.

2.3 Key Concepts

- Authenticated Landing Page: The first screen a user sees after logging in, which is customized based on their unique identity and permissions.
- Role-Aware Interface: A design approach where the information shown (such as badges and icons) changes based on the user's specific job function within the system.
- Task Summary: A high-level overview that provides immediate visibility into the user's current productivity status, specifically highlighting drafts and items requiring review.
- User Context: The collection of personal data—such as name, role, and regional settings—that allows the application to provide a relevant and localized experience.

2.4 Business Value

This feature improves user efficiency by eliminating the need to navigate through multiple menus to find pending work. By providing an immediate, role-specific summary upon login, the platform reduces cognitive load and helps users prioritize their tasks as soon as they start their session. This personalized approach ensures that every user feels recognized and informed, leading to a more intuitive and professional user experience.

3. Data

3.1 Overview

The "Onboarding" feature serves as the initial handshake between the user and the application. When a user logs in, the system automatically gathers their profile information and a summary of their current workload. This ensures that the dashboard is immediately personalized and ready for use the moment the user lands on the page, without requiring them to click through multiple screens to see their status.

3.2 Step-by-Step Flow

1. Authentication Check: The system identifies the user based on their secure login session. If the session is invalid or missing, access is denied.
2. Profile Retrieval: The system fetches the user's core details, including their name, contact information, assigned roles, and regional preferences (such as time zone and language).
3. Workload Aggregation: The system scans the user's assigned tasks and categorizes them by status (e.g., drafts, items needing review, rejected items, and approved items).
4. Dashboard Initialization: The system combines the profile data and the task counts into a single summary, which is sent to the dashboard to populate the user interface.

3.3 Key Concepts

- Session Context: The secure information that confirms who the user is, allowing the system to tailor the experience specifically to them.
- Role-Awareness: The system recognizes the user's specific permissions (e.g., Admin or Editor), which dictates what they can see and do within the application.
- Task Aggregation: A background calculation that counts a user's active work items, allowing the dashboard to display helpful "at-a-glance" status badges.
- Bootstrap: The process of loading all necessary information at once so the user experiences a fast, fully functional dashboard immediately upon arrival.

3.4 Business Value

- Improved User Experience: By providing a personalized dashboard immediately upon login, users can see exactly what requires their attention without manual searching.
- Increased Efficiency: Consolidating profile and task data into a single step reduces the number of requests the system needs to make, resulting in a faster, more responsive application.
- Operational Clarity: Users are kept informed of their workflow status (such as rejected or approved items) in real-time, helping them prioritize their daily tasks effectively.

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02 - ICF Dashboard

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Consent IQ Dashboard serves as the central command center for managing Informed Consent Forms (ICFs). It provides a high-level summary of your consent portfolio and a detailed, searchable list of all protocols, allowing users to track the status of documents and quickly navigate to specific tasks like editing or reviewing.

1.2 Step-by-Step Flow

1. Dashboard Initialization: Upon loading, the system displays a summary of your consent portfolio, including the total number of forms, those currently in review, and those already approved.
2. Search and Discovery: Users can type into the search bar to instantly filter the list of protocols by name or ICF title. If the search bar is empty, the system displays the full list of protocols.
3. Sorting: Users can click on any column header (such as Protocol ID, Version, or Last Modified) to organize the data. Clicking a header toggles the order between ascending and descending, indicated by a small arrow.
4. Managing Documents: Each protocol row provides specific actions based on the document's status:

* View: Allows users to see a preview of the document.

* Open: For drafts or documents in review, this button takes the user to the editing workspace. For approved documents, it directs the user to the final preview.

5. Creating New Content: Users can click the "Create New Consent" button at any time to begin the setup process for a new document.

1.3 Key Concepts

- Status Badge: A visual indicator that clearly labels the current stage of a document (e.g., Draft, In Review, or Approved) to help users prioritize their work.
- Dynamic Sorting: A feature that allows users to reorder the table instantly by clicking column headers, making it easier to find the most recently modified or specific protocol versions.
- Contextual Actions: The system intelligently determines which buttons to show (View vs. Open) based on the document's status, ensuring users are always directed to the correct workflow.
- Real-time Search: The dashboard updates the list of protocols as you type, removing the need to press "Enter" or click a search button.

1.4 Business Value

This dashboard streamlines the administrative burden of managing complex clinical documentation. By centralizing all protocols into one searchable, sortable view, teams can reduce the time spent hunting for documents and gain

immediate visibility into the status of their consent forms. The clear distinction between "Draft," "In Review," and "Approved" statuses helps prevent bottlenecks and ensures that users are always taking the appropriate action on the right document.

2. Logic

2.1 Overview

The ICF Dashboard serves as the central command center for managing consent forms across all research protocols. It provides users with an immediate snapshot of their workload, alongside powerful tools to organize, filter, and locate specific documents quickly. By consolidating protocol and consent form data into one view, the dashboard streamlines the administrative process for tracking document status and history.

2.2 Step-by-Step Flow

1. Dashboard Initialization: Upon loading the page, the system automatically retrieves high-level summary statistics, including the total number of protocols and a breakdown of those currently "In Review" versus "Approved."
2. Protocol Listing: The system displays a comprehensive list of all protocols, including their associated consent forms and file details.
3. Sorting: Users can organize the protocol list by clicking on column headers. The system supports toggling between ascending and descending order to help users prioritize their view.
4. Searching: As a user types into the search bar, the dashboard instantly filters the list. The search functionality scans both protocol names and consent form names simultaneously to ensure relevant results are surfaced immediately.
5. Dynamic Updates: Clearing the search input automatically restores the full, sorted list of protocols, ensuring a seamless transition between focused searching and general browsing.

2.3 Key Concepts

- Protocol: The primary research study container that holds specific consent form information and associated documentation.
- ICF (Informed Consent Form): The specific document associated with a protocol that is being tracked, reviewed, and managed within the dashboard.
- Dashboard Metrics: Real-time status counts that provide an immediate overview of the current workload and approval progress.
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Dynamic Filtering: The ability for the system to update the displayed list instantly based on user input or sorting preferences without requiring a page refresh.

2.4 Business Value

- Improved Efficiency: By providing instant search and sorting capabilities, users spend less time hunting for documents and more time managing them.
- Enhanced Visibility: The summary statistics provide leadership and administrators with an immediate pulse on the status of all consent forms, highlighting bottlenecks in the review process.
- Centralized Management: Consolidating protocol and consent form data into a single interface reduces the need to navigate between multiple systems, minimizing errors and improving data consistency.

3. Data

3.1 Overview

The ICF Dashboard serves as the central command center for managing consent forms within the BeOne system. It provides users with a high-level summary of their current workload and a detailed, searchable list of all protocols and their associated consent forms. This feature is designed to help users quickly assess the status of their projects and locate specific documentation without navigating through multiple screens.

3.2 Step-by-Step Flow

1. Dashboard Initialization: When the dashboard loads, the system automatically retrieves summary statistics, including the total number of protocols and a breakdown of those currently "In Review" versus "Approved."
2. Protocol Listing: The system displays a comprehensive list of all protocols. Users can organize this list by choosing specific sorting criteria (such as name or date) to prioritize their work.
3. Searching: Users can perform targeted searches by entering a protocol name or a specific consent form name. The system filters the entire library to display only the relevant matches.
4. Detailed View: For every protocol listed, the dashboard provides access to the associated consent form details, including version history and any linked files, ensuring all necessary information is available in one place.

3.3 Key Concepts

- Protocol: The primary research or clinical study framework that acts as the parent container for all related consent forms.
- ICF (Informed Consent Form): The specific documentation associated with a protocol, which can have multiple versions and supporting files.

- Dashboard Stats: A high-level snapshot showing the volume of work and the current progress of protocols (e.g., how many are pending review).
- Work Queue: The interactive list of protocols that allows users to sort, search, and manage their active tasks efficiently.

3.4 Business Value

- Improved Efficiency: By centralizing all protocol and consent form data, users spend less time searching for files and more time managing their research.
- Enhanced Visibility: Real-time status updates (In Review vs. Approved) allow managers to quickly identify bottlenecks and track the progress of consent documentation.
- Streamlined Workflow: The ability to sort and search through large volumes of data ensures that users can maintain control over complex projects with multiple versions and documents.

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07 - ICF Creation - Preview

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1. Frontend

1.1 Overview

The ICF Preview feature provides a final "read-only" view of the Informed Consent Form (ICF) before it is officially issued. This screen allows users to verify the document's layout, content, and structure in a simulated PDF format, ensuring everything is accurate before the final publication.

1.2 Step-by-Step Flow

1. Loading: When the user navigates to the preview page, the system automatically retrieves the document details and all associated sections, organizing them in the correct order.
2. Review: The document is displayed on a simulated PDF page. Users can view the full content, including the title, protocol metadata, and all numbered sections.
3. Adjustment: A toolbar at the top allows users to view the document filename and page count, as well as zoom in or out to inspect specific details.
4. Navigation: If the user identifies any errors, they can select the "Back to Review" button to return to the previous editing stage.
5. Publication: Once the user is satisfied, they click the "Publish" button. Upon success, the button is replaced by a confirmation badge, indicating the document is now officially published.

1.3 Key Concepts

- Simulated PDF View: A visual representation of the final document that mimics how the printed or exported file will look, allowing for a final quality check.
- Document Sections: The individual components of the ICF (such as study objectives, risks, or participant rights) that are pulled together and displayed in their specific, logical order.
- Publishing: The final action that locks the document content and marks it as officially issued within the system.
- Review Stage: A safety gate that allows users to navigate backward to correct any typographical or formatting errors before the document is finalized.

1.4 Business Value

This feature acts as the final quality assurance gate in the ICF creation process. By providing a realistic preview, it reduces the risk of errors in sensitive legal and medical documentation. It gives users the confidence that the document is formatted correctly and ready for distribution, while the "Back to Review" functionality ensures that any last-minute corrections can be made easily without disrupting the workflow.

2. Logic

2.1 Overview

The ICF (Informed Consent Form) Preview and Publication feature allows users to review a document in a simulated PDF format before finalizing it. This ensures that the document content, structure, and metadata are accurate and ready for distribution. Once the user is satisfied with the preview, they can officially publish the document, which locks the content and records the exact date and time of publication.

2.2 Step-by-Step Flow

1. Data Retrieval: When you open the preview page, the system automatically gathers all necessary information, including the document's title, version, and status, as well as the specific content sections that make up the document.
2. Document Simulation: The system uses this retrieved data to generate a simulated PDF view, allowing you to see exactly how the document will appear to the end user.
3. Final Review: You review the document to ensure all information is correct and that it has passed any required approval checks.
4. Publication: Once approved, you trigger the publish action. The system saves the document as "Published" and attaches a permanent timestamp to record when the final version was created.
5. Confirmation: Upon a successful publish, the interface updates to display a confirmation badge, indicating the document is now live and finalized.

2.3 Key Concepts

- ICF (Informed Consent Form): The core document being managed, which requires a formal review and publication process.
- Preview: A simulated view of the document that allows users to verify the layout and content before it is officially released.
- Metadata: The background information about the document, such as its title, current version, and status, which helps identify and track the file.
- Publication: The final step in the document lifecycle that locks the content and marks it as official, accompanied by a timestamp for audit purposes.

2.4 Business Value

This feature provides a reliable "what-you-see-is-what-you-get" experience, reducing the risk of errors in sensitive consent documentation. By automating the publication timestamp and locking the document upon release, the system ensures data integrity and maintains a clear, auditable history of when each version of a document was finalized.

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05 - ICF Creation - Edit and Refine

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1. Frontend

1.1 Overview

The "Edit and Refine" feature provides a user-friendly interface for reviewing and polishing AI-generated Informed Consent Form (ICF) documents. It allows users to view the document section-by-section, make manual text adjustments, and trigger AI-assisted rewrites to ensure the content meets specific requirements.

1.2 Step-by-Step Flow

1. Document Loading: Upon opening the editor, the system retrieves the document and displays it in a structured layout, including a table of contents and a progress tracker.
2. Section Review: The document is broken down into individual "Section Cards." Each card displays the section title and number. If a section was originally created by AI, it is clearly marked with an "AI Generated" badge.
3. Initiating Refinement: To modify a section, the user clicks the "Edit with AI" button on the specific section card.
4. Manual Editing & AI Assistance: A pop-up window appears, displaying the current text. The user can manually edit this text directly. Once satisfied with the manual changes, the user can trigger the AI to further refine or rewrite the content.
5. Processing Feedback: While the AI is processing a request, the section card displays a loading overlay to inform the user that an update is in progress.

1.3 Key Concepts

- Section Card: A modular container that displays a specific part of the document, allowing for focused review and individual editing.
- AI-Assisted Rewrite: A feature that uses artificial intelligence to improve, clarify, or rephrase the text within a section based on the user's input.
- Manual Override: The ability for users to directly edit any section text, ensuring they maintain full control over the final document content.
- Loading Overlay: A visual indicator that appears during AI processing to prevent duplicate actions and provide status updates.

1.4 Business Value

- Efficiency: Reduces the time required to finalize complex legal and medical documents by automating the drafting and refinement process.
- Accuracy & Control: Empowers users to combine the speed of AI generation with the precision of human oversight, ensuring the final document is both compliant and high-quality.

- User Experience: Provides a clean, organized interface that breaks down long, dense documents into manageable pieces, reducing cognitive load for the reviewer.

2. Logic

2.1 Overview

The "Edit and Refine" feature provides a dedicated workspace for reviewing and polishing AI-generated Informed Consent Forms (ICFs). When you open a document, the system automatically retrieves the latest version of the content and its associated metadata. This ensures that you are always working with the most up-to-date information, allowing you to seamlessly transition between viewing the document structure and making manual or AI-assisted refinements.

2.2 Step-by-Step Flow

1. Document Initialization: When you navigate to the edit view, the system automatically requests the specific document by its unique identifier.
2. Data Retrieval: The system fetches the full document package, which includes the document name, current status, version history, and the actual text content.
3. Loading State: While the system gathers this information, a loading indicator is displayed to ensure you know the document is being prepared for your review.
4. Display: Once the data is retrieved, the document is rendered in the center panel, organizing the content into distinct sections that are ready for your review, manual edits, or AI-assisted updates.

2.3 Key Concepts

- Document Metadata: Information about the document, such as its name, version number, and current status, which helps you track the progress of the ICF.
- Content Retrieval: The process of pulling the latest version of the document text from the system so that your edits are applied to the most recent draft.
- Section-Based Editing: A layout approach where the document is broken down into manageable parts, allowing you to focus on specific areas for refinement without needing to manage the entire document at once.

2.4 Business Value

This feature streamlines the document review process by providing a reliable, "single source of truth" for your ICFs. By automating the retrieval and organization of document sections, it reduces the administrative burden on users,

minimizes the risk of working on outdated drafts, and provides a clear, intuitive interface for finalizing high-quality consent forms efficiently.

3. Data

3.1 Overview

The "Edit and Refine" feature provides a user-friendly interface for managing Informed Consent Form (ICF) documents. It allows users to review AI-generated content, make manual adjustments, and leverage AI assistance to rewrite specific sections of the document. This ensures that the final document is accurate, compliant, and tailored to the specific needs of the study.

3.2 Step-by-Step Flow

1. Document Retrieval: When you open a document, the system fetches the full document details, including its current content, version information, and status, to display it in the editor.
2. Manual Editing: You can directly modify the text within the document. The system validates your changes to ensure the document remains in a valid state and saves your updates automatically.
3. AI-Assisted Refinement: If you need help with a specific section, you can select that section and provide instructions (a prompt) to the AI.
4. Content Update: The AI generates a refined version of that specific section based on your instructions, which is then returned to the editor for your review and final approval.
5. Persistence: Every time a change is made—whether manual or AI-assisted—the system updates the document record and logs the time of the last modification.

3.3 Key Concepts

- Manual Edit: The ability for a user to type directly into the document to correct or update information.
- AI-Assisted Rewrite: A feature that uses artificial intelligence to rephrase or improve specific parts of the document based on user-provided guidance.
- Document State: The current version of the ICF, including its text, metadata, and the timestamp of the most recent change.
- Section Management: The capability to target specific parts of the document for editing or AI refinement without affecting the entire document.

3.4 Business Value

This feature significantly reduces the administrative burden of drafting ICFs by combining the speed of AI generation with the precision of human oversight. By allowing for both manual control and AI-powered suggestions,

the tool ensures high-quality, compliant documentation while drastically shortening the time required to finalize complex consent forms.

BUSINESS REQUIREMENTS DOCUMENT

03 - ICF Creation - Setup

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1. Frontend

1.1 Overview

The ICF Creation Setup feature is a three-column wizard designed to help users prepare the necessary information for generating an Informed Consent Form (ICF). By guiding the user through selecting a clinical protocol, attaching essential reference documents, and choosing a document template, the system ensures that the AI has all the required context to produce an accurate and compliant document.

1.2 Step-by-Step Flow

1. Protocol Selection: The user searches for and selects a specific clinical protocol from the available list. This selection acts as the foundation for the rest of the setup.
2. Reference Attachment: Once a protocol is selected, the user provides the required reference materials—specifically an "Investigation Brochure" and a "Lab Manual." These can be uploaded directly from the user's device or fetched from the Document Management System (DMS).
3. Template Selection: The user chooses a document template to define the structure of the ICF. A "standard" template is selected by default, but users can choose others based on their specific needs.
4. Launch Generation: Once a protocol and template are selected, the "Launch" button becomes active, allowing the user to trigger the AI generation process.

1.3 Key Concepts

- Protocol: The clinical study framework that provides the core data for the ICF.
- Reference Documents: Supporting files (Investigation Brochure and Lab Manual) that provide the AI with the necessary background information to generate accurate content.
- Template: A pre-defined document structure that ensures the generated ICF meets organizational formatting and content standards.
- DMS (Document Management System): A centralized repository where existing reference documents can be retrieved without needing to upload them manually.
- Wizard Interface: A multi-step layout that breaks down a complex task into manageable, sequential actions to prevent user error.

1.4 Business Value

This feature streamlines the document creation process by centralizing all necessary inputs into a single, intuitive interface. By requiring users to link specific protocols and reference documents before generation, the system significantly reduces the risk of errors and ensures that every ICF is generated with the correct, up-to-date clinical

context. This automation saves time for clinical teams and improves the consistency and quality of regulatory documentation.

2. Logic

2.1 Overview

The ICF (Informed Consent Form) Creation Setup feature is a three-step wizard designed to prepare the necessary information for AI-driven document generation. Users guide the system by selecting a specific clinical protocol, attaching relevant reference documents (such as an Investigation Brochure or Lab Manual), and choosing a document template. This structured approach ensures the AI has the correct context and formatting requirements before it begins drafting the ICF.

2.2 Step-by-Step Flow

1. Protocol Selection: Upon opening the wizard, the system loads a list of available protocols. Users can browse the list or use the search bar to find a specific protocol by name or ID.
2. Reference Document Attachment: Once a protocol is selected, the user attaches required reference documents. For each document category (e.g., Investigation Brochure), the user has two options:

* Fetch from DMS: Retrieve an existing document already stored in the system database associated with that protocol.

* Upload: Upload a new file directly from their local device.

3. Template Selection: The user selects an ICF template from a provided list. The system highlights recommended templates to help guide the user toward the best choice for their specific needs.
4. Validation & Readiness: Throughout the process, the system validates that a protocol is selected before allowing document attachments. Once all components are configured, the setup is complete and ready for the AI generation phase.

2.3 Key Concepts

- Protocol: The clinical study framework that defines the scope and requirements for the ICF.
- Reference Documents: Supporting materials (like brochures or manuals) that provide the AI with the necessary background information to generate an accurate and compliant ICF.
- Template: A pre-defined structure or "blueprint" that dictates the layout and standard language of the generated ICF.
- DMS (Document Management System): The central repository where existing reference documents are stored and can be retrieved for use in the wizard.

2.4 Business Value

This feature streamlines the ICF creation process by centralizing all necessary inputs into a single, intuitive interface. By allowing users to easily link protocols with specific reference materials and templates, the system reduces manual data entry errors and ensures that the AI generation engine receives high-quality, relevant context. This ultimately accelerates the document drafting timeline while maintaining consistency across clinical study documentation.

3. Data

3.1 Overview

The ICF (Informed Consent Form) Creation Setup feature provides a streamlined, three-step wizard designed to help users prepare for AI-assisted document generation. By guiding the user through selecting a clinical protocol, managing necessary reference documents, and choosing a document template, the system ensures that all required information is organized and ready for the generation process.

3.2 Step-by-Step Flow

1. Protocol Selection: Upon opening the wizard, the system displays a list of available clinical protocols. Users can browse the full list or use a search function to quickly find a specific protocol by its name or ID.
2. Reference Document Management: Once a protocol is selected, the user can view existing reference documents associated with that protocol. If additional files are needed, the user can upload them directly into the system, which automatically links the files to the chosen protocol.
3. Template Selection: The user selects an appropriate ICF template from a curated list. The system highlights recommended templates to assist the user in making the best choice for their specific needs.
4. Preparation Completion: With the protocol, reference documents, and template confirmed, the setup is complete, and the system is ready to launch the AI generation process.

3.3 Key Concepts

- Protocol: The core clinical study framework that defines the context for the document being generated.
- Reference Document: Supporting files (such as study summaries or safety information) that provide the necessary data for the AI to create an accurate ICF.
- Template: A pre-defined document structure that ensures the generated ICF meets organizational standards and formatting requirements.
- Wizard: A user-friendly interface that breaks a complex task into manageable, sequential steps to reduce errors and improve efficiency.

3.4 Business Value

- Efficiency: By centralizing protocol selection, document uploads, and template choices in one wizard, users save time and avoid switching between different systems.
- Accuracy: Linking reference documents directly to a specific protocol ensures that the AI uses the correct, context-specific information, leading to higher-quality document drafts.
- Standardization: Using pre-approved templates ensures that all generated ICFs remain consistent with company policies and regulatory expectations.
- User Experience: The intuitive, step-by-step design lowers the barrier to entry for new users and reduces the cognitive load required to initiate complex document generation tasks.

06 - ICF Creation - Review

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1. Frontend

1.1 Overview

The ICF Review feature provides a centralized workspace for stakeholders to evaluate draft Informed Consent Forms (ICFs). It allows reviewers to read the document, view feedback, track the history of changes, and perform final approval or rejection actions. The interface is designed to provide transparency by clearly distinguishing between human-authored content and AI-generated sections.

1.2 Step-by-Step Flow

1. Document Loading: Upon opening the review page, the system retrieves the full document content and the complete history of actions taken on that document.
2. Navigation and Context: The user is presented with a header and a progress tracker to maintain context regarding the document's protocol and current stage in the workflow.
3. Document Review: The document is displayed in a structured format, with each section presented as an individual card. Sections generated by AI are clearly labeled with a visual badge.
4. Feedback and History: A side panel provides access to comments and a chronological "Audit Trail," which lists every significant action (such as edits or previous reviews) taken on the document.
5. Decision Making: Using the review actions bar, the reviewer can compare different versions of the document to see what has changed and then formally approve or reject the draft.

1.3 Key Concepts

- Audit Trail: A comprehensive, time-stamped log of all activities related to the document, ensuring full accountability and traceability for compliance.
- Section Cards: A modular way of displaying the document, where each part of the ICF is broken down into a readable, organized block.
- AI-Generated Badge: A visual indicator that helps reviewers quickly identify which parts of the document were drafted by AI, allowing them to focus their scrutiny appropriately.
- Version Comparison: A tool that allows reviewers to see the differences between document drafts, making it easier to verify that feedback has been correctly incorporated.

1.4 Business Value

This feature streamlines the document approval process by providing a single source of truth for reviewers. By integrating the audit history, comment threads, and version comparison tools directly into the review interface, the system reduces the time spent switching between different platforms. Furthermore, the clear labeling of

AI-generated content and the detailed audit trail support regulatory compliance and quality assurance, ensuring that all changes are transparent and documented.

2. Logic

2.1 Overview

The ICF Review feature provides a dedicated workspace for stakeholders to examine draft Informed Consent Forms (ICFs). This interface allows reviewers to view the current version of a document alongside a complete history of all actions taken on it. By combining the document content with a transparent audit trail, the system ensures that reviewers have the necessary context to provide feedback and track the document's progress toward finalization.

2.2 Step-by-Step Flow

1. Document Loading: When a user opens the Review Page, the system retrieves the full document, including its current content, version information, and status.
2. Content Display: The document is rendered in the center panel, allowing the reviewer to read the text and evaluate the draft.
3. Audit Trail Retrieval: Simultaneously, the system fetches a chronological list of all activities associated with that specific document.
4. History Presentation: The audit trail is displayed to the user, showing a clear record of who performed specific actions and when those actions occurred.

2.3 Key Concepts

- ICF (Informed Consent Form): The core document being reviewed, which contains the specific protocol details and consent language for a study.
- Audit Trail: A chronological, tamper-evident log that tracks the lifecycle of the document, including review comments, status changes, and user interactions.
- Document Metadata: Supporting information about the file, such as its version number, creation date, and last modification timestamp, which helps reviewers verify they are looking at the correct draft.

2.4 Business Value

- Improved Accuracy: By providing a clear view of the document and its history, reviewers can easily identify previous feedback and ensure that requested changes have been incorporated.
- Regulatory Compliance: The detailed audit trail provides full traceability, which is essential for maintaining compliance and accountability throughout the document approval process.

- Efficiency: Centralizing the document content and the history of actions in one view reduces the time spent searching for context, allowing for a faster and more informed review cycle.

3. Data

3.1 Overview

The ICF Review feature allows users to access a draft version of an Informed Consent Form (ICF) to evaluate its content. It provides a dedicated workspace where reviewers can see the current document alongside a history of previous actions and feedback. This ensures that the document can be refined iteratively based on reviewer comments until it is ready for finalization.

3.2 Step-by-Step Flow

1. Accessing the Review Workspace: When a user opens a document for review, the system retrieves the current version of the ICF content and its associated metadata.
2. Viewing the Audit Trail: Simultaneously, the system pulls a chronological list of all past actions taken on that document. This history is displayed from newest to oldest, allowing the reviewer to quickly catch up on recent changes or feedback.
3. Reviewing and Incorporating Feedback: The reviewer examines the document content and the activity history to identify necessary adjustments.
4. Updating the Document: Once the reviewer makes the required changes, the system saves the updated content and metadata. The document's "last modified" timestamp is automatically updated to reflect that a new iteration has been created.

3.3 Key Concepts

- Audit Trail: A chronological record of all activities performed on a document, helping users track the history of changes and feedback.
- Document Metadata: Descriptive information about the ICF, such as its version number and status, which helps keep the document organized.
- Iterative Updates: The ability to modify and save the document multiple times, ensuring that feedback is captured and reflected in the latest version.

3.4 Business Value

This feature streamlines the document approval process by centralizing the review and editing tasks. By providing a clear history of feedback, it reduces the risk of miscommunication between stakeholders and ensures that all

requested changes are accurately incorporated. This leads to higher quality documentation and a more efficient path toward finalizing the ICF.

BUSINESS REQUIREMENTS DOCUMENT

04 - ICF Creation - Generate

Prepared by Kolosal AI

Version 1.0 **Date** October 20, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The ICF Generation feature provides a transparent, real-time view of the document creation process. When the system begins generating an Informed Consent Form (ICF), users are presented with a dedicated dashboard that tracks progress through six specific stages. This ensures users are kept informed about the status of their request without needing to guess how much time remains or what the system is currently doing.

1.2 Step-by-Step Flow

1. Initiation: Once the generation process starts, the system displays a progress dashboard showing the specific protocol being processed.
2. Real-Time Monitoring: The screen updates automatically every second, showing a progress bar and a countdown timer (ETA) to help manage expectations.
3. Stage Tracking: A checklist of six sequential steps is displayed. As the system works, each step transitions from "Pending" to "Active" (indicated by a spinner) and finally to "Done" (indicated by a green checkmark).
4. Completion: Once the final step is finished and the progress reaches 100%, the system automatically redirects the user to the document editor to review the newly created ICF.

1.3 Key Concepts

- Progress Visualization: A visual representation of the generation status, including a progress bar and a step-by-step checklist, to provide clarity on the system's current activity.
- Dynamic ETA: A countdown timer that estimates the remaining time until the document is ready, helping users understand how long they need to wait.
- Automated Navigation: A seamless transition that moves the user from the waiting screen directly into the document editor as soon as the file is ready, removing the need for manual intervention.
- Generation Pipeline: The six-stage sequence the system follows, ranging from parsing protocol data to performing final compliance checks.

1.4 Business Value

This feature significantly improves the user experience by replacing uncertainty with transparency. By providing a clear, step-by-step view of the generation process, users can trust that the system is actively working on their request. The automated navigation to the editor upon completion saves time and reduces friction, allowing users to move immediately from the generation phase to the review and editing phase without unnecessary clicks or delays.

2. Logic

2.1 Overview

The ICF Generation feature provides a transparent, real-time view of the document creation process. Instead of leaving you on a static loading screen, the system visually tracks the AI's progress through six distinct stages. Once the document is fully generated, the system automatically transitions you to the editor so you can begin reviewing and refining your content without needing to click any additional buttons.

2.2 Step-by-Step Flow

1. Initialization: As soon as the generation process begins, the system establishes a live connection to track the progress of your document.
2. Real-Time Updates: As the AI works, the screen updates dynamically to show a progress bar and highlights which of the six sequential steps is currently being processed.
3. Status Monitoring: You will see a "Generating your ICF..." message while the work is in progress.
4. Completion: Once the AI finishes the final step, the status updates to "Finalizing..." to indicate the document is ready.
5. Auto-Navigation: After a brief pause to ensure the document is fully prepared, the system automatically redirects you to the editor screen to view your new document.

2.3 Key Concepts

- Progress Visualization: A visual interface that breaks down the complex generation process into six easy-to-understand steps, keeping you informed of exactly what the system is doing.
- Live Stream: A continuous connection between the server and your screen that allows the progress bar and status labels to update instantly without you needing to refresh the page.
- Auto-Navigation: A seamless transition feature that detects when the document is ready and moves you directly to the next phase of your workflow, reducing manual effort.

2.4 Business Value

This feature improves the user experience by removing the uncertainty often associated with AI-driven tasks. By providing clear, step-by-step feedback, users can trust that the system is actively working. Furthermore, the automatic transition to the editor streamlines the workflow, allowing users to move immediately from document creation to document refinement, thereby increasing overall productivity.

BUSINESS REQUIREMENTS DOCUMENT

Global Navigation Header

Prepared by Kolosal AI

Version 1.0 **Date** October 20, 2026 **Status** Draft **Classification** Internal

1. components

1.1 Overview

The Global Navigation Header is a persistent, fixed bar that appears at the top of every page within the Consent IQ platform. It serves as the primary orientation point for users, providing clear branding, platform identification, and quick access to personal account information and system notifications.

1.2 Step-by-Step Flow

1. System Initialization: As the page loads, the system automatically attempts to retrieve the current user's information and notification status from the platform's database.
2. Data Retrieval: The system fetches the user's name, assigned role, and the number of unread notifications. If the system cannot reach the database, it automatically applies default placeholder values to ensure the header remains visible and functional.
3. Brand Display: The left side of the header displays the "Consent IQ" platform name, a descriptive subtitle, and the "Powered by Agilisium" attribution.
4. Notification Alerting: The right side of the header displays a bell icon. If there are unread notifications, a small badge appears over the icon indicating the count.
5. User Identity Display: The right side also displays the user's profile information, including an avatar (represented by the user's initials), their full name, and their current role, accompanied by a dropdown menu icon for further account actions.

1.3 Key Concepts

- Persistent Header: A navigation bar that stays locked at the top of the screen, ensuring users can always access their profile or notifications regardless of where they are in the application.
- Graceful Fallback: A safety mechanism that ensures the header displays default information (such as "ICF Author") if the system encounters a connection error, preventing the interface from breaking.
- Notification Badge: A visual indicator that provides immediate feedback on pending tasks or alerts without requiring the user to navigate away from their current page.
- User Context: The specific data set (name, role, and initials) that personalizes the user's experience within the platform.

1.4 Business Value

- Improved User Experience: By providing a consistent navigation anchor, users can move through the platform intuitively without losing their place.
-

Operational Continuity: The fallback logic ensures that even if backend services experience temporary issues, the user is not blocked from using the platform, maintaining productivity.

- Brand Recognition: The prominent display of the platform name and attribution reinforces the professional identity of the Consent IQ tool.
- Efficiency: The notification badge allows users to stay informed about important updates at a glance, reducing the need to manually check for new information.

2. Data

2.1 Overview

The Global Navigation Header is a consistent top-bar component designed to provide users with immediate access to their identity and status within the platform. It ensures that regardless of which page a user is on, they can see their name, professional role, and a summary of their current workload. The system is designed to be resilient, providing sensible default information if a user's profile data is temporarily unavailable.

2.2 Step-by-Step Flow

1. The system identifies the currently logged-in user based on their active session.
2. The system retrieves the user's profile details, including their name, primary and secondary roles, and the number of tasks currently assigned to them.
3. If the system successfully finds the user, it displays their specific information in the header.
4. If the system cannot find the user (or if data is missing), it automatically displays a default profile: "User Name" with a role of "ICF Author" and zero tasks.
5. The header displays this information alongside a dropdown menu icon, allowing the user to interact with their profile settings.

2.3 Key Concepts

- User Context: The collection of personal details (name, roles, and task volume) that the system uses to personalize the navigation experience.
- Resilience (Fallback Logic): A safety mechanism that ensures the navigation header remains functional and professional even if the system encounters an error or cannot retrieve specific user data.
- Task Summary: A quick-reference indicator that shows the user how many items currently require their attention.

2.4 Business Value

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Improved User Experience: By providing a consistent header across the platform, users always know who is logged in and can easily access their profile.

- Operational Efficiency: Displaying task counts directly in the header helps users stay informed about their workload without needing to navigate to a separate dashboard.
- System Reliability: The fallback mechanism ensures that the interface remains clean and usable even during unexpected data issues, preventing broken layouts or confusing empty states for the end user.

BUSINESS REQUIREMENTS DOCUMENT

Progress Stepper

Prepared by Kolosal AI

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1. Components

1.1 Overview

The Progress Stepper is a visual navigation tool displayed at the top of the creation screens in the BeOne platform. It provides users with a clear sense of where they are in the process and how much work remains. The stepper guides users through five distinct stages: Setup, Generate, Edit & Refine, Review, and Preview.

1.2 Step-by-Step Flow

1. Visual Tracking: As you move through the creation process, the stepper automatically updates to show your current location, which steps you have already finished, and which steps are still ahead.
2. Status Indicators: Each step is visually represented based on its status:

* Done: Steps you have completed are marked with a check icon.

* Active: The step you are currently working on is highlighted to stand out.

* Future: Steps you have not yet reached are grayed out.

3. Navigation: You can click on previously completed steps to return to them, provided the system has enough information (such as a saved document ID) to load that specific page.
4. System Constraints: The "Generate" step is a temporary, automated process and cannot be clicked to navigate back to. Additionally, later stages like "Edit & Refine," "Review," and "Preview" require a saved document to be accessible.

1.3 Key Concepts

- Active Step: The specific stage of the process currently displayed on your screen.
- Completed Steps: Stages that have been finished, allowing you to revisit them if needed.
- Future Steps: Stages that are locked until you progress further through the workflow.
- Transient Step: A step (like "Generate") that acts as a bridge between stages and does not serve as a destination you can navigate to manually.
- Document ID: A unique identifier required to ensure that when you navigate back to a previous step, the system loads the correct work-in-progress.

1.4 Business Value

The Progress Stepper improves the user experience by providing transparency and predictability. By clearly showing the path from start to finish, it reduces user confusion, helps manage expectations regarding the length of the process, and allows for easy navigation back to previous steps to make adjustments without losing context.

BUSINESS REQUIREMENTS DOCUMENT

Wrapper Shell

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1. components

1.1 Overview

The Wrapper Shell is a consistent layout component designed for the BeOne POC creation flow. It acts as a "frame" that surrounds the main content of each page, ensuring that users always have access to essential navigation tools and context about the specific protocol they are working on. By standardizing this layout, we ensure a uniform experience across all five steps of the consent creation process.

1.2 Step-by-Step Flow

1. Global Navigation: The system displays a top bar that provides consistent access to global navigation and user identity information.
2. Contextual Breadcrumbs: A sub-bar appears below the top bar, showing the user's path. This includes a link back to the main dashboard, a static label identifying the current process ("Consent Creation"), and dynamic details about the specific protocol (such as its ID, name, and version) if that information is available.
3. Progress Tracking: A visual step indicator is displayed to show the user exactly where they are within the five-step creation flow.
4. Content Display: The main area of the screen renders the specific content for the current step, framed by the navigation and progress elements above.

1.3 Key Concepts

- Wrapper Shell: A structural container that provides a consistent look and feel across different pages of the application.
- Breadcrumb Sub-bar: A navigation aid that helps users understand their location within the system and provides quick access to parent pages and specific protocol details.
- Conditional Rendering: A logic feature where specific information (like protocol ID or version) only appears on the screen if that data is currently available for the user.
- Step Indicator: A visual tool that tracks progress through a multi-step process, helping users understand how much of the task is complete.

1.4 Business Value

- Improved User Orientation: By providing constant access to breadcrumbs and progress indicators, users are less likely to feel lost while navigating the multi-step consent creation process.
- Consistency: Standardizing the layout reduces cognitive load, as users do not have to re-learn the navigation structure when moving between different steps of the workflow.
- Efficiency: The dynamic display of protocol information ensures that users always have the necessary context (such as the specific protocol ID or version) visible without needing to navigate away from their current task.

BUSINESS REQUIREMENTS DOCUMENT

Table of Contents

Prepared by Kolosal AI

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1. components

1.1 Overview

The Table of Contents (TOC) feature provides a navigation sidebar on the left side of the document view. It acts as a roadmap for the document, allowing users to quickly jump to specific sections while maintaining a clear view of the document's structure. By clicking an item in the TOC, the main document automatically scrolls to the corresponding section, and the right-hand panel updates to show relevant details for that specific part of the document.

1.2 Step-by-Step Flow

1. Organization: The system analyzes the document sections and organizes them into a list. It automatically detects sub-sections (e.g., "3.1") to ensure they are visually indented and properly categorized under main headings.
2. Display: The TOC is rendered in the left panel, showing a numbered list of all document sections.
3. Selection: When a user clicks on a section title in the TOC, the system highlights that item to indicate it is the "active" section.
4. Navigation: The main document view smoothly scrolls to the selected section so it is centered for the user.
5. Context Update: Simultaneously, the system triggers an update to the right-hand panel, loading the specific details or AI-assisted tools associated with the selected section.

1.3 Key Concepts

- Hierarchical Rendering: The ability to distinguish between main sections and sub-sections, displaying them with appropriate visual hierarchy (indentation) to make the document easier to scan.
- Active State Management: A visual indicator that tracks which section the user is currently viewing, ensuring the user always knows their location within the document.
- Synchronized Navigation: The link between the TOC and the main document, where a single click performs two actions: moving the user to the correct place in the text and updating the side panel with relevant information.

1.4 Business Value

- Improved Efficiency: Users no longer need to manually scroll through long documents to find specific information, saving time during review.
- Enhanced User Experience: By providing a persistent navigation menu, the interface feels more intuitive and professional, reducing the cognitive load on the user.
-

Contextual Awareness: By automatically updating the right-hand panel when a section is selected, the system ensures that users always have the correct tools and information available for the specific section they are working on.

2. LOGIC

2.1 Overview

The Table of Contents feature provides a navigation sidebar that allows users to easily view, manage, and jump to specific parts of a document. It acts as a dynamic map of the document's structure, ensuring that as you add or organize content, the navigation panel stays updated and synchronized with your current view.

2.2 Step-by-Step Flow

1. Initial Load: When you open a document, the system automatically retrieves all existing sections and displays them in the left-hand panel, ordered by their position in the document.
2. Active Tracking: As you scroll through the document or select a specific heading, the Table of Contents highlights the corresponding section to show you exactly where you are.
3. Adding Content: When you click "Add Section" or "Add Subsection," the system creates a new entry in the document.
4. Automatic Refresh: Immediately after adding a new section, the Table of Contents updates to include the new item.
5. Seamless Navigation: The system automatically selects the newly created section and scrolls your view directly to it, allowing you to begin editing immediately.

2.3 Key Concepts

- Section: A distinct block of content within the document that appears as an entry in the Table of Contents.
- Position: The specific order in which a section appears, ensuring the Table of Contents matches the document flow.
- Active Selection: The visual indicator in the sidebar that tells you which part of the document is currently being viewed or edited.
- Dynamic Refresh: The ability of the sidebar to update itself automatically whenever the document structure changes, without requiring a manual page reload.

2.4 Business Value

This feature significantly improves document readability and user efficiency. By providing a persistent navigation aid, users can quickly orient themselves in long or complex documents. Furthermore, the ability to instantly add

and jump to new sections reduces the time spent manually scrolling or searching, allowing for a more fluid and productive content creation experience.

3. DATA

3.1 Overview

The Table of Contents (TOC) feature provides a dynamic navigation panel on the left side of the document interface. It allows users to view the structure of their document at a glance and easily manage its organization. The system automatically tracks the order of sections, ensuring that the TOC always reflects the current layout of the document as users add or update content.

3.2 Step-by-Step Flow

1. Loading the Document: When a user opens a document, the system retrieves the full list of sections associated with that document.
2. Ordering: The system automatically sorts these sections based on their assigned position, ensuring the TOC displays them in the correct sequence from top to bottom.
3. Adding New Content: When a user clicks "Add Section" or "Add Subsection," the system creates a new entry.
4. Automatic Positioning: The system automatically assigns the new section a position number (placing it at the end of the current list) so the user does not have to manually manage the document's order.
5. Display Update: The TOC refreshes to include the new section, allowing the user to navigate to it immediately.

3.3 Key Concepts

- Section: A distinct block of content within a document that appears as an entry in the Table of Contents.
- Positioning: An automated sorting mechanism that ensures sections appear in the TOC in the exact order they are intended to be read.
- Dynamic Refresh: The ability of the TOC to update instantly whenever a new section is added, keeping the navigation panel in sync with the document content.

3.4 Business Value

- Improved User Experience: Users can quickly jump to specific parts of a long document without scrolling, saving time and reducing frustration.
- Reduced Administrative Burden: By automating the numbering and ordering of sections, the system eliminates manual formatting tasks, allowing users to focus on the document content.
-

Consistency: Automated positioning ensures that the document structure remains organized and predictable, which is critical for maintaining professional standards in complex documentation.

BUSINESS REQUIREMENTS DOCUMENT

AI Document Editor

Prepared by Kolosal AI

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1. COMPONENTS

1.1 Overview

The AI Document Editor is a feature within the "Edit and Refine" page that allows users to interact with AI to improve document content. Users can manually edit text directly or provide specific instructions to the AI to rewrite, simplify, or summarize sections. The tool is designed to be intuitive, offering quick-start prompts to help users get the best results from the AI with minimal effort.

1.2 Step-by-Step Flow

1. Review and Edit: The user views the document section and can manually make changes to the text in the editor.
2. Provide Instructions: The user either types a custom request into the instruction box or selects a "Quick Prompt" (such as "Simplify language for patient readability") to automatically populate the instruction box.
3. Trigger AI: Once an instruction is provided, the user clicks the "Ask AI" button.
4. Save and Process: The system automatically saves any manual changes made to the text, sends the content and the user's instruction to the AI service, and displays an "Asking..." status while waiting for the result.
5. Update: Once the AI completes the task, the document text is instantly updated with the new version, providing immediate feedback to the user.

1.3 Key Concepts

- Quick Prompts: Pre-written commands that save time and guide the user on how to best utilize the AI for common tasks like simplifying language.
- Instruction Input: A flexible text area where users can provide natural-language requests to guide the AI's editing style or focus.
- Manual Override: The ability for users to edit text manually before or after AI intervention, ensuring they always maintain final control over the document.
- State Synchronization: A background process that ensures the text the user sees in the editor matches the official document record, preventing data loss during AI processing.

1.4 Business Value

- Efficiency: Reduces the time spent on manual drafting and refining by automating common editing tasks.
- Consistency: Ensures document quality by providing standardized options for simplifying complex language or generating summaries.
-

User Empowerment: Gives non-technical users an easy-to-use interface to leverage powerful AI capabilities without needing to understand the underlying technology.

- Data Integrity: By automatically saving manual changes before running AI tasks, the system ensures that no user work is lost during the editing process.

2. Logic

2.1 Overview

The AI Document Editor feature allows you to refine specific sections of your document using natural language instructions. Instead of manually rewriting text, you can provide a prompt—such as "simplify this language" or "check for compliance"—and the AI will automatically update the content of that section while maintaining the document's structure.

2.2 Step-by-Step Flow

1. Selection and Instruction: You select a specific section of your document and enter a natural language instruction into the AI prompt field.
2. Context Gathering: The system retrieves the current content and title of the selected section to ensure the AI understands exactly what it is working on.
3. AI Processing: The system sends your instruction and the existing section content to the AI service, which generates a refined version of the text based on your request.
4. Update and Versioning: The system replaces the old content with the new, AI-generated version. It also increments the version number of that section to keep an accurate audit trail of all changes made.
5. Display: The updated content is immediately reflected in your document editor, allowing you to review the changes instantly.

2.3 Key Concepts

- Natural Language Prompt: A simple, plain-English instruction you provide to tell the AI how you want the text changed (e.g., "make this more professional").
- Contextual Awareness: The AI's ability to see the existing section content and title, ensuring that its suggestions are relevant and accurate to your specific document.
- Version Control: An automatic tracking system that assigns a new version number every time the AI modifies a section, ensuring you can always track the history of edits.
- Refinement: The process of using AI to improve, rewrite, or adjust existing text based on user-defined goals.

2.4 Business Value

- Increased Efficiency: Reduces the time spent on manual drafting and editing by automating repetitive writing tasks.
- Improved Quality: Ensures consistent tone, clarity, and compliance across documents by leveraging AI to apply professional standards.
- Seamless Workflow: Allows users to stay within the document editor while performing complex edits, preventing the need to switch between different tools or applications.
- Auditability: Maintains a clear history of document changes, providing peace of mind and transparency throughout the editing process.

3. Data

3.1 Overview

The AI Document Editor feature allows users to refine specific sections of a document using artificial intelligence. When a user provides a prompt to improve or rewrite a section, the system automatically processes the request, updates the content, and maintains a clear audit trail of the changes. This ensures that users can easily iterate on their documents while keeping track of which parts have been modified by AI.

3.2 Step-by-Step Flow

1. Request Initiation: The user submits a request to refine a specific document section by providing a prompt (instructions on how to change the text) and the current content of that section.
2. AI Processing: The system sends the user's prompt and the existing text to the AI engine, which generates the refined version of the content.
3. Version Tracking: The system retrieves the current version of the section and increments the version number to ensure a history of changes is maintained.
4. Update and Labeling: The system updates the section with the new content, tags it as "AI Edited," and sets the status to "AI Generated."
5. Persistence: The updated section is saved to the database, and the new version is returned to the user for review.

3.3 Key Concepts

- Refinement Prompt: The specific instructions provided by the user to guide the AI in rewriting or improving the document text.
- Version Control: A system that automatically increments a counter every time a section is updated, allowing users to track the evolution of their document.

- Provenance Tagging: A labeling system that identifies whether a section was modified by AI or manually, providing transparency regarding the source of the content.
- Status Tracking: A workflow state that indicates the current condition of a section (e.g., "AI Generated"), helping users manage the document review process.

3.4 Business Value

- Increased Efficiency: Users can quickly rewrite or polish complex document sections without manual drafting, significantly reducing the time spent on editing.
- Improved Accuracy and Consistency: By using AI to refine text, users can ensure that document sections remain clear, professional, and aligned with organizational standards.
- Auditability: The automatic versioning and labeling features provide a clear history of changes, which is essential for maintaining document integrity and compliance in professional environments.

BUSINESS REQUIREMENTS DOCUMENT

AI Openrouter SDK

Prepared by Kolosal AI

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1. FRONTEND

1.1 Overview

The AI Document Generation feature allows users to create professional documents automatically by combining pre-defined templates with specific source data. By leveraging the OpenRouter AI service, the system intelligently fills in document variables and structures content, significantly reducing the manual effort required to draft and format complex files.

1.2 Step-by-Step Flow

1. Data Entry: The user provides the necessary information to start the process, including a specific template identifier, the name of the source file containing the raw data, and any additional custom values needed for the document.
2. Validation: The system checks that all required information is present to ensure the AI has everything it needs to generate an accurate result.
3. AI Processing: The system sends the template and source data to the OpenRouter AI service. The AI analyzes the structure and content to generate the final document.
4. Result Delivery: Once the AI completes the generation, the system displays the final document, including the total page count and the formatted content, for the user to review.

1.3 Key Concepts

- Template ID: A unique reference code that tells the system which document layout or structure to use as a foundation.
- Source Filename: The specific file containing the raw information that the AI will use to populate the template.
- Values: Custom data points or variables that allow the user to personalize the document beyond the standard template content.
- OpenRouter SDK: The underlying AI technology that interprets the instructions and source data to perform the actual writing and formatting of the document.

1.4 Business Value

- Efficiency: Automates the time-consuming task of manually drafting and formatting documents, allowing staff to focus on higher-value activities.
- Consistency: Ensures that all generated documents follow a standardized structure and professional quality, reducing the risk of human error.
-

Scalability: Enables the rapid creation of large volumes of documents, making it easier to handle increased workloads without needing additional manual resources.

2. LOGIC

2.1 Overview

The AI OpenRouter SDK feature enables the system to automatically generate and structure documents by leveraging advanced Artificial Intelligence models. By connecting to the OpenRouter platform, the system can take a source file (such as a PDF) and specific user-provided data to create a new, formatted document. This process not only generates the content but also intelligently organizes it into logical sections for easier management and retrieval.

2.2 Step-by-Step Flow

1. File Preparation: The system retrieves the requested source document and converts it into a digital format that the AI can read and process.
2. Context Assembly: The system combines the source document with specific instructions (the template ID and user-provided data) to create a clear request for the AI.
3. AI Processing: This request is sent to an AI model via OpenRouter, which analyzes the file and the instructions to generate the new document content.
4. Data Storage: Once the AI returns the generated content, the system saves the full document into the database for future reference.
5. Structural Parsing: The system automatically scans the generated content to identify logical headers and sections, saving each part individually so that users can interact with specific sections of the document rather than just the whole file.

2.3 Key Concepts

- OpenRouter: A service that provides access to various high-performance AI models through a single, unified connection.
- Template: A predefined structure or set of rules that tells the AI how to format and organize the information provided by the user.
- Base64 Encoding: A method used to convert a file (like a PDF) into a text-based format so it can be safely transmitted to the AI service.
- Markdown: A lightweight formatting language used by the AI to structure the generated text (e.g., bolding, lists, and headers).

- Persistence: The process of saving the AI's output into our database, ensuring that generated documents are stored securely and remain available for later use.

2.4 Business Value

- Efficiency: Automates the time-consuming task of drafting documents from scratch, allowing staff to focus on higher-value activities.
- Consistency: Ensures that all generated documents follow the same professional structure and formatting standards defined by your templates.
- Improved Accessibility: By breaking documents into structured sections, the system makes it easier for users to navigate, edit, or extract specific information from long or complex files.
- Flexibility: By using OpenRouter, the system can easily switch between different AI models to find the best balance of cost, speed, and quality without requiring major software changes.

3. DATA

3.1 Overview

The AI OpenRouter SDK feature enables our system to automatically process and generate documents by leveraging advanced AI models. By connecting to the OpenRouter platform, the system can ingest PDF files, analyze their content, and produce structured, high-quality document outputs that are saved directly into our database for future use.

3.2 Step-by-Step Flow

1. File Preparation: The system takes a source PDF document and converts it into a digital format (base64) that the AI can read and process.
2. AI Analysis: The system sends the document and specific instructions to an AI model via the OpenRouter interface.
3. Content Generation: The AI analyzes the document and generates the requested content, which is returned to our system.
4. Record Creation: The system creates a primary record in the database to store metadata about the document, such as the template used, the model selected, and the total page count.
5. Structured Storage: The system breaks the generated content into individual, organized sections (e.g., titles and body text) and saves them into the database, ensuring the information is easy to retrieve and manage later.

3.3 Key Concepts

- OpenRouter: A service that provides access to various powerful AI models through a single, unified connection.
- Base64: A standard method for encoding files into text, allowing documents like PDFs to be sent securely to AI services.
- Document Generation: The process of using AI to create or fill out document templates based on provided source material.
- Structured Data: Organizing the AI's output into specific sections (like titles and content) rather than one long block of text, which makes the information more useful for our applications.

3.4 Business Value

- Efficiency: Automates the time-consuming task of manually extracting and formatting information from complex documents.
- Consistency: Ensures that generated documents follow a standardized structure and quality, regardless of the source material.
- Scalability: By using a flexible AI integration, we can easily swap or upgrade models as technology improves without needing to rebuild our entire document processing workflow.
- Accessibility: Storing content in a structured database format allows other parts of our software to easily search, display, or report on the generated document data.

BUSINESS REQUIREMENTS DOCUMENT

Edit Actions

Prepared by Kolosal AI

Version 1.0 **Date** October 20, 2026 **Status** Draft **Classification** Internal

1. COMPONENTS

1.1 Overview

The Edit Actions Bar is a control panel designed for the "Edit and Refine" page. It provides users with a clear, consistent interface to manage specific sections of a document. By centralizing essential actions like editing, deleting, and previewing, the bar ensures that users can modify document content or verify their changes without navigating away from their current workspace.

1.2 Step-by-Step Flow

1. Context Display: The bar automatically displays the current document name, version, and the title of the section you are currently working on.
2. Action Selection:

* To Edit: Click "Edit" to switch the section into an active text-entry mode. You can then "Save" your changes to update the document or "Cancel" to discard them.
* To Delete: Click "Delete" to trigger a confirmation window. This prevents accidental removals by requiring a second step to finalize the deletion.
* To Preview: Click "Preview Draft" to open a full-screen view of the document. This allows you to see how your changes look in the final layout before finalizing your work.

3. Completion: Once an action (Save or Delete) is successfully processed, the system refreshes the view to reflect the updated document state.

1.3 Key Concepts

- Edit Mode: A state that transforms a static section into an interactive text area, allowing for direct content modification.
- Confirmation Modal: A safety feature that appears before permanent actions (like deleting) to ensure the user intends to proceed.
- Preview Overlay: A full-screen, read-only view that renders the document as a PDF, providing a "final look" at the document structure including any recent AI-driven updates.
- State Management: The underlying logic that ensures the interface only shows relevant buttons (e.g., hiding "Edit" while you are already in the middle of editing).

1.4 Business Value

- Efficiency: Reduces the time spent navigating between different pages by providing all necessary document management tools in one place.

- Data Integrity: The use of confirmation modals and a "Cancel" option during editing helps prevent accidental data loss or unintended changes to critical documents.
- User Confidence: The "Preview" feature allows users to verify the impact of their edits in a professional, document-ready format, ensuring high-quality output before finalizing the document.

2. logic

2.1 Overview

The Edit Actions feature provides users with the necessary tools to manage document content directly within the "Edit and Refine" page. This feature ensures that users can easily modify text or remove sections while maintaining document integrity, audit trails, and proper organization.

2.2 Step-by-Step Flow

1. Document Loading: When the "Edit and Refine" page opens, the system retrieves the full document, including its metadata, status, and all existing sections, to display them in the center panel.
2. Editing Content: When a user chooses to edit a section, the system switches to an input mode. Once the user saves their changes, the system updates the section content, refreshes the document's "last modified" timestamp, and creates an audit log entry to track the manual change.
3. Deleting Sections: If a user chooses to delete a section, the system displays a confirmation prompt to prevent accidental data loss. Upon confirmation, the section is removed, the remaining sections are automatically re-sequenced to maintain the correct order, and the document's "last modified" timestamp is updated.
4. ID Generation: For new documents, the system automatically generates a unique identifier using a standardized format (e.g., PROTO-2026-782-ICF) to ensure consistency and prevent duplicate entries.

2.3 Key Concepts

- Audit Logging: A background process that records every manual edit made to a document, ensuring there is a clear history of changes for compliance and tracking.
- Re-sequencing: An automated process that adjusts the order of remaining document sections after one has been deleted, ensuring the document structure remains logical and unbroken.
- Metadata: Essential information about the document, such as its name, version, status, and modification history, which is loaded automatically to provide context to the user.
- Confirmation Prompt: A safety mechanism that requires the user to verify their intent before performing a permanent action, such as deleting a section.

2.4 Business Value

- Data Integrity: By automatically re-sequencing sections and updating timestamps, the system ensures the document remains organized and accurate without requiring manual intervention from the user.
- Compliance and Accountability: The automatic audit logging of manual edits provides a transparent trail of changes, which is critical for regulated document workflows.
- User Efficiency: Providing direct "Edit" and "Delete" actions within the document view reduces the need to navigate away from the page, streamlining the refinement process.
- Standardization: The automated generation of unique ICF IDs eliminates the risk of duplicate identifiers and ensures that all documents follow a consistent naming convention.

3. data

3.1 Overview

The Edit Action Bar is a core component of the "Edit and Refine" page, providing users with the tools necessary to manage document content. It allows users to view document details, modify specific sections of text, and remove sections that are no longer needed. The system is designed to ensure that every change is tracked for auditing purposes and that the document structure remains organized even after deletions.

3.2 Step-by-Step Flow

1. Loading the Document: When a user opens the "Edit and Refine" page, the system retrieves the full document, including its metadata, status, and current content, to display it in the center panel.
2. Updating Content: When a user makes manual text changes to a specific section, the system saves the new content and automatically creates an entry in the activity log to record that a manual edit occurred.
3. Deleting a Section: If a user chooses to remove a section, the system deletes it from the document.
4. Re-sequencing: Immediately following a deletion, the system automatically updates the order of the remaining sections to ensure the document maintains a correct, sequential structure.

3.3 Key Concepts

- Document Metadata: The background information associated with a document, such as its name, version, status, and creation/modification timestamps.
- Activity Log: A secure record that tracks specific user actions (like manual edits) to provide an audit trail for document changes.
- Section Sequencing: The logical order of parts within a document; the system ensures this order is preserved and corrected automatically whenever a section is removed.
- Manual Edit: Any direct change made by a user to the text content of a document section.

3.4 Business Value

- Data Integrity: By automatically re-sequencing sections after a deletion, the system prevents broken document structures and ensures the document remains professional and readable.
- Accountability: The automatic activity logging provides a clear history of changes, which is essential for compliance and tracking the evolution of a document.
- User Efficiency: The Edit Action Bar centralizes essential document management tasks, allowing users to refine content quickly without needing to manually reorganize the document or track their own edit history.

BUSINESS REQUIREMENTS DOCUMENT

Review Actions

Prepared by Kolosal AI

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1. COMPONENTS

1.1 Overview

The Review Actions Bar is a navigation and utility tool located at the top of the document review page. Its primary purpose is to provide clear context about the document you are currently viewing and to allow you to easily compare different versions of that document side-by-side. This ensures that reviewers can quickly identify changes between iterations without leaving their current workspace.

1.2 Step-by-Step Flow

1. Context Display: When you open the review page, the bar automatically displays the document name and the specific version you are currently viewing (e.g., "v1.0 Draft").
2. Accessing Comparison: You click the "Compare Versions" button to view a list of all available versions for that document.
3. Version Selection: A dropdown menu appears, showing all historical versions of the document, including their status and labels.
4. Side-by-Side View: Once you select a different version from the list, the system automatically opens a split-screen view, placing your current version next to the selected version for easy comparison.

1.3 Key Concepts

- Document Context: The persistent display of the document name and current version, ensuring you always know exactly which file iteration you are working on.
- Version History: A comprehensive list of all previous iterations of a document, allowing for historical tracking and auditability.
- Side-by-Side Comparison: A specialized view mode that aligns two versions of a document simultaneously, making it easier to spot edits, additions, or deletions.

1.4 Business Value

- Efficiency: Reduces the time spent manually searching for previous versions or switching between different browser tabs to compare document changes.
- Accuracy: Minimizes the risk of human error during the review process by providing a clear, visual comparison of document iterations.
- Clarity: Provides immediate transparency into the document's lifecycle, helping stakeholders understand the progression of drafts and approvals at a glance.

2. Logic

2.1 Overview

The "Compare Versions" feature in the Review Action Bar allows users to easily navigate and evaluate the history of a document. By providing a clear list of all available versions, users can quickly switch between different iterations to see how a document has evolved over time. This feature ensures that reviewers always have the context of previous changes at their fingertips.

2.2 Step-by-Step Flow

1. Trigger: The user clicks the "Compare Versions" button within the Review Action Bar.
2. Data Retrieval: The system automatically identifies the current document and fetches the complete history of all associated versions.
3. Processing: The system organizes the version data, including the version number, its current status, and whether it is the most recent version.
4. Display: A dropdown menu appears, presenting the list of versions clearly so the user can select one for comparison.
5. Comparison: Upon selecting a different version, the system displays that version side-by-side with the current document for easy review.

2.3 Key Concepts

- Version History: A chronological record of all saved iterations of a document, allowing users to track changes.
- Status Badge: A visual indicator that shows the current state of a specific version (e.g., draft, pending, or approved).
- Current Version Indicator: A label that highlights which version is the most up-to-date, helping users distinguish it from historical drafts.
- Side-by-Side Comparison: A viewing mode that places two versions of a document next to each other to make identifying differences straightforward.

2.4 Business Value

This feature significantly improves the efficiency and accuracy of the document review process. By enabling side-by-side comparisons, it eliminates the need for manual cross-referencing, reduces the risk of human error, and provides reviewers with the necessary transparency to make informed decisions based on the document's evolution.

BUSINESS REQUIREMENTS DOCUMENT

Comment Actions

Prepared by Kolosal AI

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1. components

1.1 Overview

The "Comments & Approval" feature provides a centralized workspace on the document review page for team collaboration. It allows users to discuss specific sections of a document, track the status of feedback, and ensures that all necessary reviews and resolutions are completed before a document can be officially submitted for approval.

1.2 Step-by-Step Flow

1. Reviewing Comments: Users can view a scrollable list of all comments. Each comment displays the relevant document section, the author's details, the timestamp, and the message.
2. Managing Conversations: Users can reply to existing comments or update the status of a comment (e.g., marking it as resolved) directly within the comment card.
3. Adding Feedback: Users can add new comments using the pinned composer at the bottom of the panel, ensuring each comment is linked to a specific section of the document.
4. Tracking Reviewer Progress: The "Approval Gate" at the bottom of the panel tracks which reviewers have completed their tasks. Reviewers can click an "Act" button to update their status.
5. Final Submission: Once all comments are marked as resolved and all reviewers have completed their actions, the "Submit for Approval" button becomes active, allowing the document to move forward in the workflow.

1.3 Key Concepts

- Comment Threading: A way to organize discussions by allowing users to reply to specific comments, keeping the conversation focused and readable.
- Status Toggle: A control on each comment that allows team members to change the state of a discussion (e.g., from "Open" to "Resolved").
- Approval Gate: A safety mechanism that prevents a document from being submitted until specific criteria—such as zero open comments and full reviewer participation—are met.
- Pinned Composer: A fixed input area at the bottom of the panel that allows users to quickly add new feedback without scrolling.

1.4 Business Value

This feature streamlines the document review process by eliminating ambiguity and preventing premature submissions. By requiring all comments to be resolved and all reviewers to participate before submission, the system

ensures high-quality output and accountability. The centralized interface reduces the time spent switching between different tools, allowing teams to collaborate efficiently and maintain a clear audit trail of all feedback and decisions.

2. Logic

2.1 Overview

The Comment Actions feature provides a dedicated space on the document review page for team members to communicate, track feedback, and manage the approval process. It allows users to view threaded conversations, post new comments or replies, and update their individual review status, ensuring that all feedback is addressed before a document is submitted for final approval.

2.2 Step-by-Step Flow

1. Loading Comments: When a user opens the document, the system automatically retrieves all existing comments and replies, displaying them in a threaded list within the right-hand panel.
2. Adding Feedback: Users can add new comments or reply to existing ones using the pinned composer at the bottom of the panel. Once submitted, the new entry is saved and immediately appears in the thread.
3. Managing Review Status: Within the approval area, reviewers can click an "Act" button to confirm their participation. The system updates their status to "acted" in real-time.
4. Approval Validation: The system continuously monitors two conditions: that all comments are marked as "Resolved" or "Closed," and that every reviewer has performed their "Act" action.
5. Submission: Once all comments are resolved and all reviewers have acted, the "Submit for Approval" button becomes active, allowing the user to finalize the document review.

2.3 Key Concepts

- Threaded Comments: A structured view where replies are nested under their original parent comment, keeping conversations organized and easy to follow.
- Reviewer Status: A tracking mechanism that records whether a specific reviewer has completed their required actions on the document.
- Approval Gate: A safety check that prevents a document from being submitted until all feedback is addressed and all reviewers have signed off.
- Real-time Updates: The ability for the comment thread to refresh instantly when a new comment is added, without requiring the user to manually reload the page.

2.4 Business Value

This feature streamlines the review cycle by centralizing communication and accountability. By enforcing an "Approval Gate," it ensures that no document is submitted prematurely, reducing the risk of overlooked feedback or incomplete reviews. The real-time nature of the comment thread fosters faster collaboration, allowing teams to resolve issues efficiently and move projects forward with confidence.

3. Data

3.1 Overview

The Comment Actions feature provides a dedicated space on the document review page for users to collaborate. It allows reviewers to post comments, reply to existing discussions, and track the resolution status of specific feedback. Additionally, the system automatically tracks when a reviewer has taken action on a document, ensuring clear visibility into the review progress.

3.2 Step-by-Step Flow

1. Viewing Comments: When a user opens a document, the system automatically retrieves all existing comments and replies associated with that specific document and displays them in the right-hand panel.
2. Adding Feedback: A user can create a new comment or reply to an existing one. The system saves this input, assigns it an "open" status, and updates the view in real-time.
3. Managing Status: As discussions progress, users can update the status of a comment (e.g., marking an "open" issue as "resolved") to help the team track what still requires attention.
4. Tracking Reviewer Activity: Once a reviewer interacts with the document, the system updates their personal status to "acted," providing a clear indicator that they have participated in the review process.

3.3 Key Concepts

- Comment Threading: The ability to group original comments with their subsequent replies, keeping related feedback organized.
- Comment Status: A label assigned to each comment (such as "open" or "resolved") that helps teams manage the lifecycle of feedback.
- Reviewer Activity Tracking: A system-level flag that records whether a specific reviewer has performed an action on the document, helping project leads monitor participation.
- Real-time Updates: The system’s ability to display new comments or status changes immediately without requiring the user to refresh the page.

3.4 Business Value

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Improved Collaboration: By centralizing feedback in a dedicated panel, team members can discuss specific document sections directly, reducing the need for external communication tools.

- Enhanced Accountability: Tracking reviewer activity ensures that project managers can easily identify who has completed their review and who is still pending.
- Streamlined Workflow: The ability to mark comments as "resolved" allows teams to focus on outstanding issues, ensuring that no critical feedback is overlooked during the review cycle.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Edit

Prepared by Kolosal AI

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1. components

1.1 Overview

The Bottom Action Bar is a navigation and control component located at the bottom of the "Edit and Refine" page. It provides users with a clear way to either return to the previous configuration step or finalize their work by submitting the document for formal review.

1.2 Step-by-Step Flow

1. User Interaction: The user is presented with two options: "Back to Setup" or "Submit for Review."
2. Navigation (Back to Setup): If the user selects "Back to Setup," the system immediately redirects them to the initial configuration page for the current protocol.
3. Submission (Submit for Review): If the user selects "Submit for Review," the system locks the button to prevent duplicate requests while it processes the action.
4. Processing: The system sends the document to the review queue.
5. Outcome:

* If the submission is successful, the user is automatically navigated to the review page.

* If an error occurs, the system displays a notification to the user explaining that the submission failed, and the button becomes active again so the user can try again.

1.3 Key Concepts

- Ghost Button: A secondary button style used for "Back to Setup" that is visually less prominent than the primary action, helping users focus on the main task.
- Primary Action: The "Submit for Review" button, which serves as the main call-to-action for completing the editing phase.
- Submission State: A safety feature that disables the "Submit for Review" button while the system is working, preventing the user from accidentally sending multiple requests.
- Document Lifecycle: The process of moving a document from a "Draft" state to an "In Review" state, ensuring that work is formally handed off for approval.

1.4 Business Value

This feature streamlines the user experience by providing consistent, predictable navigation during the document creation process. By clearly separating the "Setup" and "Review" phases, it reduces user error and ensures that documents are only moved to the review stage when the user is ready, maintaining data integrity and workflow efficiency.

2. logic

2.1 Overview

The "Submit for Review" feature allows users to finalize their work on the "Edit and Refine" page. By clicking this button, the user signals that the document is ready for the next stage of the workflow, moving it from a draft state into an official review process.

2.2 Step-by-Step Flow

1. Initiation: The user clicks the "Submit for Review" button located in the bottom action bar.
2. Verification: The system checks to ensure the document exists and is valid for submission.
3. Status Update: The system automatically updates the document's status from "Draft" to "In Review" and records the time of the update.
4. Confirmation: The system confirms the update and prepares the document for the next phase of the workflow.

2.3 Key Concepts

- Document Status: A label that tracks where a document is in its lifecycle (e.g., moving from a working draft to a formal review).
- In Review: The specific status assigned to a document once the user has finished editing and submitted it for approval.
- Workflow Transition: The automated process of moving a document from one stage to another, ensuring the system always reflects the current state of the work.

2.4 Business Value

This feature streamlines the transition between editing and approval, reducing manual administrative work. By automating the status change, it ensures that stakeholders are immediately notified that a document is ready for review, which helps maintain project momentum and ensures that documents move through the pipeline efficiently without unnecessary delays.

3. Data

3.1 Overview

The "Edit and Refine" page features a bottom action bar designed to help users manage their documents efficiently. This interface provides two primary functions: saving ongoing work as a draft and submitting a completed document

for formal approval. The system ensures that documents move through a controlled workflow, preventing accidental submissions and maintaining clear status tracking.

3.2 Step-by-Step Flow

1. Saving Progress: When a user makes changes to a document, they can save their work. The system updates the document content and metadata while keeping the document in its current state.
2. Initiating Submission: When the user is ready to finalize their work, they trigger the submission action from the bottom bar.
3. Status Verification: The system automatically checks the document's current status. It will only proceed if the document is currently marked as a "Draft."
4. Finalizing Submission: If the document is a valid draft, the system updates its status to "In Review," signaling that it is ready for the next stage of the process. If the document is not in a draft state, the system will prevent the submission to avoid errors.

3.3 Key Concepts

- Draft: The initial state of a document where the user is actively editing content.
- In Review: The status assigned to a document once it has been submitted, indicating it is no longer editable by the author and is awaiting further action.
- Status Transition: The controlled movement of a document from one stage (Draft) to another (In Review) to ensure data integrity.
- Metadata: Additional descriptive information attached to the document that is saved alongside the content during updates.

3.4 Business Value

- Data Integrity: By enforcing status checks, the system prevents users from accidentally submitting documents that are not ready or are already in the review process.
- Workflow Efficiency: The bottom action bar provides a clear, consistent user experience, allowing authors to easily toggle between saving their progress and moving their work forward.
- Audit Readiness: Every update and status change is tracked with a timestamp, ensuring there is a clear record of when a document was modified or submitted for review.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Setup

Prepared by Kolosal AI

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1. COMPONENTS

1.1 Overview

The "Bottom Bar - Setup" feature provides a persistent navigation and action area at the bottom of the Setup page. It is designed to guide users through the final steps of preparing their document generation by ensuring all necessary requirements are met before proceeding. It also offers a quick way to return to the main dashboard at any time.

1.2 Step-by-Step Flow

1. Navigation Options: Users can click the "Back to Dashboard" button at any time to exit the setup process and return to the main dashboard.
2. Validation: The system continuously monitors the setup page to ensure three specific requirements are met: a protocol is selected, at least one reference is attached, and a template is selected.
3. Action Activation: Once all three requirements are satisfied, the "Generate ICF" button automatically becomes active.
4. Initiating Generation: When the user clicks "Generate ICF," the system triggers the document generation process and immediately moves the user to the progress screen.
5. Progress Tracking: On the progress screen, the user can view a six-step tracker that displays the real-time status of the generation. Once the process is complete, the system automatically redirects the user to the document editor.

1.3 Key Concepts

- Conditional Activation: A safety feature that prevents the generation process from starting until all required information (protocol, references, and template) is provided.
- Persistent Action Bar: A fixed interface element at the bottom of the screen that ensures navigation and primary actions are always accessible to the user.
- Generation Pipeline: The automated background process that takes the selected inputs and creates the ICF draft.
- Progress Visualization: A status display that provides transparency into the generation process, showing the user exactly which stage the system is currently completing.

1.4 Business Value

This feature streamlines the user experience by providing clear, guided steps for document creation. By enforcing validation before generation, it reduces the risk of errors and ensures that the system only processes complete

requests. The persistent navigation and real-time progress tracking keep the user informed and in control, ultimately increasing efficiency and confidence in the document generation workflow.

2. logic

2.1 Overview

The "Generate ICF" feature, located in the bottom action bar of the Setup page, allows users to initiate the creation of an Informed Consent Form (ICF) draft. By selecting a protocol and a template, the system automatically extracts and structures the necessary information from your documents to prepare the draft for further processing.

2.2 Step-by-Step Flow

1. Initiation: Once you have selected a protocol, attached the required references, and chosen a template, the "Generate ICF" button becomes active.
2. Request Submission: Clicking the button sends your request to the system to begin the generation process.
3. Job Creation: The system creates a new "Generation Job" to track the progress of your request and immediately confirms that the process has started.
4. Document Parsing: In the background, the system retrieves the relevant document, extracts the text, and organizes it into a structured format.
5. Downstream Processing: Once the document is successfully parsed, the structured data is automatically passed to the next stage of the pipeline to finalize the ICF draft.

2.3 Key Concepts

- Generation Job: A tracking record created by the system to manage your request from start to finish, ensuring you know the current status of your document generation.
- Protocol: The specific clinical study or research framework that provides the source material for the ICF.
- Template: A pre-defined layout or structure that dictates how the final ICF document should be formatted.
- Parsing: The automated process of reading a raw document and identifying its individual sections and data points so the system can use that information to build the new form.

2.4 Business Value

This feature streamlines the document creation process by removing the need for manual data entry. By automating the extraction and structuring of information from existing protocols, the system reduces the risk of human error, ensures consistency across all generated forms, and significantly accelerates the time it takes to prepare an ICF draft for review.

3. Data

3.1 Overview

The "Setup" page bottom action bar includes a feature that allows users to initiate the generation of an Informed Consent Form (ICF). When a user triggers this action, the system begins a background process to create the document, allowing the user to continue their work without waiting for the file to finish generating.

3.2 Step-by-Step Flow

1. Initiation: The user clicks the generate action on the Setup page.
2. Validation: The system checks the provided information to ensure it is complete and accurate.
3. Job Creation: The system creates a new "pending" task in the background to handle the document generation.
4. Confirmation: The system immediately sends a confirmation back to the user, providing a unique job ID that identifies this specific generation request.
5. Background Processing: The system works in the background to finalize the document, keeping the user interface responsive.

3.3 Key Concepts

- ICF Generation: The automated process of creating an Informed Consent Form based on the specific protocol and template selected by the user.
- Background Processing: A method where the system performs complex tasks "behind the scenes," ensuring the user does not have to wait on a loading screen.
- Job ID: A unique reference number assigned to each generation request, which allows the system to track the progress of that specific document.
- Pending Status: The initial state of a request, indicating that the system has received the instruction and is preparing to process it.

3.4 Business Value

This feature improves user productivity by removing the need to wait for document generation to complete before moving to the next task. By handling the process in the background, the application remains fast and responsive, providing a smoother experience for users managing complex protocol setups.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Review

Prepared by Kolosal AI

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1. Components

1.1 Overview

The "Review" page features a persistent bottom action bar designed to give users clear control over their document workflow. This bar provides a quick way to either return to the editing phase if further changes are needed or to finalize the document by submitting it for approval. It also provides real-time feedback regarding any outstanding comments that may need attention before submission.

1.2 Step-by-Step Flow

1. Initialization: When the user arrives at the Review page, the system automatically checks the document status to see if there are any unresolved comments.
2. Navigation Option: The user can select "Back to Edit" at any time to return to the previous screen to make further adjustments to the document.
3. Submission Check: The "Submit for Approval" button dynamically updates its label based on the number of open comments. If comments exist, the button displays the count (e.g., "Submit for Approval (3 open)") to alert the user.
4. Submission Action: When the user clicks "Submit for Approval," the system processes the request. During this time, the button is disabled to prevent duplicate submissions.
5. Completion: Upon a successful submission, the user is automatically redirected to a success page. If an error occurs, the user is notified so they can try again.

1.3 Key Concepts

- Dynamic Labeling: The button text changes automatically based on the document's current state, ensuring the user is always aware of pending tasks (open comments) before finalizing.
- Submission Guard: A safety feature that disables the submit button while a request is in progress, preventing accidental double-submissions or system errors.
- Workflow Navigation: A clear path back to the editing stage, allowing for a seamless transition between refining content and final approval.

1.4 Business Value

This feature improves document quality and user efficiency by providing a "final check" mechanism. By highlighting open comments directly on the submission button, it reduces the risk of submitting incomplete work. Furthermore, the clear navigation options ensure that users can move fluidly through the document lifecycle without getting stuck, ultimately leading to a faster and more reliable approval process.

2. Logic

2.1 Overview

The "Review" page features a bottom action bar designed to help users finalize their work. This bar provides a clear path to submit a document for approval while ensuring that all necessary feedback has been addressed before the document moves to the final preview stage.

2.2 Step-by-Step Flow

1. Monitor Feedback: The system automatically tracks the number of unresolved comments on the document.
2. Display Status: The "Submit for Approval" button updates its label based on the feedback status. If there are outstanding comments, the button displays the specific count to alert the user.
3. Initiate Submission: When the user clicks the button, the system triggers the approval workflow.
4. Finalize and Transition: Once the submission is successfully processed, the system automatically navigates the user to the "ICF Creation - Preview" page to view the final document.

2.3 Key Concepts

- Approval Gate: A safety check that ensures a document cannot be submitted for approval until all open comments have been resolved.
- Dynamic Labeling: A feature that provides immediate visibility into the document's readiness by showing the number of open comments directly on the action button.
- Workflow Automation: The seamless transition from the review phase to the preview phase once the submission criteria are met.

2.4 Business Value

This feature improves document quality and process efficiency by preventing premature submissions. By clearly highlighting unresolved comments, it encourages users to address feedback before moving forward, reducing the likelihood of errors and minimizing the need for rework during the final approval stages.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Generate

Prepared by Kolosal AI

Version 1.0	**Date** October 20, 2026	**Status** Draft	**Classification** Internal

1. COMPONENTS

1.1 Overview

The "Generate" page features a fixed navigation bar at the bottom of the screen designed to give users control over their workflow. This bar provides two clear options that allow users to either return to the initial configuration phase or bypass the current generation process to move directly into the editing phase.

1.2 Step-by-Step Flow

1. User Interaction: The user views the "Generate" page and decides to change their project settings or move forward.
2. Option A (Back to Setup): If the user selects "Back to Setup," the system immediately redirects them to the initial configuration page to adjust their project parameters.
3. Option B (Skip): If the user selects "Skip," the system bypasses the current generation process and takes the user directly to the editor for their specific project.

1.3 Key Concepts

- Back to Setup: A navigation option that allows users to restart or modify their project configuration if they are not satisfied with their current path.
- Skip: A shortcut that allows users to bypass the generation phase and proceed directly to the editing tools for their specific job.
- Fixed Navigation: A persistent bar at the bottom of the screen that ensures users always have access to these primary navigation choices, regardless of where they are on the page.

1.4 Business Value

This feature improves user efficiency by providing flexible navigation paths. By allowing users to easily return to the setup phase or skip ahead to editing, we reduce friction and prevent users from feeling "stuck" in the generation process. This creates a more intuitive and responsive experience, ensuring users can manage their workflow according to their own pace and needs.